Building quantum codes with transversal CCZ

from projective geometry and SAT solvers

Bohan Lu, Duke University

Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

October 22, 2026

arXiv:2610.09341

Outline

1. **Quantum error correction and transversal gates**
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Pauli strings and stabilizer codes

Binary vectors $c, c' \in \mathbb{F}_2^n$: weight $|c|$, XOR $c \oplus c'$ (not a direct sum), and overlap $c \wedge c'$ (entrywise product).

$$c \cdot c' = |c \wedge c'| \bmod 2$$

Pauli strings $X(a) = \bigotimes_j X^{a_j}$ and $Z(b) = \bigotimes_j Z^{b_j}$ on the basis $|c\rangle$:

$$X(a)|c\rangle = |c \oplus a\rangle, \qquad Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle.$$

Pauli strings and stabilizer codes

Binary vectors $c, c' \in \mathbb{F}_2^n$: weight $|c|$, XOR $c \oplus c'$ (not a direct sum), and overlap $c \wedge c'$ (entrywise product).

$$c \cdot c' = |c \wedge c'| \bmod 2$$

Pauli strings $X(a) = \bigotimes_j X^{a_j}$ and $Z(b) = \bigotimes_j Z^{b_j}$ on the basis $|c\rangle$:

$$X(a)|c\rangle = |c \oplus a\rangle, \qquad Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle.$$

They commute or anticommute:

$$X(a)Z(b) = (-1)^{a \cdot b} Z(b)X(a).$$

Pauli strings and stabilizer codes

Binary vectors $c, c' \in \mathbb{F}_2^n$: weight $|c|$, XOR $c \oplus c'$ (not a direct sum), and overlap $c \wedge c'$ (entrywise product).

$$c \cdot c' = |c \wedge c'| \bmod 2$$

Pauli strings $X(a) = \bigotimes_j X^{a_j}$ and $Z(b) = \bigotimes_j Z^{b_j}$ on the basis $|c\rangle$:

$$X(a)|c\rangle = |c \oplus a\rangle, \qquad Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle.$$

They commute or anticommute:

$$X(a)Z(b) = (-1)^{a \cdot b} Z(b)X(a).$$

Stabilizer code: the joint $+1$ eigenspace of an abelian group of Pauli strings, with the flipped eigenvalues as *syndrome*.

Pauli strings and stabilizer codes

Binary vectors $c, c' \in \mathbb{F}_2^n$: weight $|c|$, XOR $c \oplus c'$ (not a direct sum), and overlap $c \wedge c'$ (entrywise product).

$$c \cdot c' = |c \wedge c'| \bmod 2$$

Pauli strings $X(a) = \bigotimes_j X^{a_j}$ and $Z(b) = \bigotimes_j Z^{b_j}$ on the basis $|c\rangle$:

$$X(a)|c\rangle = |c \oplus a\rangle, \qquad Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle.$$

They commute or anticommute:

$$X(a)Z(b) = (-1)^{a \cdot b} Z(b)X(a).$$

Stabilizer code: the joint $+1$ eigenspace of an abelian group of Pauli strings, with the flipped eigenvalues as *syndrome*.

$[\![n, k, d]\!]$: dimension 2^k, and d is the least weight of a nontrivial *logical operator*.

CSS codes

X-type and Z-type stabilizers from $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$ [CS96, Ste96]:

$$X(a),\ a \in \mathcal{C}_2, \qquad Z(b),\ b \in \mathcal{C}_1^{\perp}.$$

Stabilizer rows S_l $(s \times n)$ and logical rows K_i $(k \times n)$:

$$\mathcal{C}_2 = \operatorname{rowspan} S, \qquad \mathcal{C}_1 = \operatorname{rowspan} G, \qquad G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right].$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes

X-type and Z-type stabilizers from $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$ [CS96, Ste96]:

$$X(a),\ a \in \mathcal{C}_2, \qquad Z(b),\ b \in \mathcal{C}_1^{\perp}.$$

Stabilizer rows S_l $(s \times n)$ and logical rows K_i $(k \times n)$:

$$\mathcal{C}_2 = \operatorname{rowspan} S, \qquad \mathcal{C}_1 = \operatorname{rowspan} G, \qquad G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right].$$

The logical basis is a uniform superposition over the coset $K^\top x \oplus \mathcal{C}_2$:

$$|x\rangle_L = 2^{-s/2} \textstyle\sum_{y \in \mathbb{F}_2^s} |K^\top x \oplus S^\top y\rangle, \qquad x \in \mathbb{F}_2^k.$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes

X-type and Z-type stabilizers from $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$ [CS96, Ste96]:

$$X(a),\ a \in \mathcal{C}_2, \qquad Z(b),\ b \in \mathcal{C}_1^{\perp}.$$

Stabilizer rows S_l ($s \times n$) and logical rows K_i ($k \times n$):

$$\mathcal{C}_2 = \operatorname{rowspan} S, \qquad \mathcal{C}_1 = \operatorname{rowspan} G, \qquad G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right].$$

The logical basis is a uniform superposition over the coset $K^{\top}x \oplus \mathcal{C}_2$:

$$|x\rangle_L = 2^{-s/2} \textstyle\sum_{y \in \mathbb{F}_2^s} |K^{\top}x \oplus S^{\top}y\rangle, \qquad x \in \mathbb{F}_2^k.$$

Logical operators $X(a)$, $a \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and $Z(b)$, $b \in \mathcal{C}_2^{\perp} \setminus \mathcal{C}_1^{\perp}$:

$$d_X = \min_{v \in \mathcal{C}_1 \setminus \mathcal{C}_2} |v|, \qquad d_Z = \min_{v \in \mathcal{C}_2^{\perp} \setminus \mathcal{C}_1^{\perp}} |v|, \qquad d = \min(d_X, d_Z).$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

Clifford and non-Clifford gates

On basis states, with $x, x_i \in \mathbb{F}_2$:

$$H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$$
$$T|x\rangle = e^{i\pi x/4}|x\rangle, \quad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

S, T and CCZ are *diagonal*.

Clifford and non-Clifford gates

On basis states, with $x, x_i \in \mathbb{F}_2$:

Clifford: $H|x\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$

non-Clifford: $T|x\rangle = e^{i\pi x/4}|x\rangle, \quad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$

S, T and CCZ are *diagonal*.

The *Clifford group* maps Pauli strings to Pauli strings under conjugation, and H, S, CNOT generate it.

Clifford and non-Clifford gates

On basis states, with $x, x_i \in \mathbb{F}_2$:

$$\text{Clifford:}\quad H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$$

$$\text{non-Clifford:}\quad T|x\rangle = e^{i\pi x/4}|x\rangle, \quad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

S, T and CCZ are *diagonal*.

The *Clifford group* maps Pauli strings to Pauli strings under conjugation, and H, S, CNOT generate it.

Clifford gates are not universal. Clifford $+\,T$ and Clifford + CCZ are. A quantum computer needs both kinds as logical gates.

Transversal gates and magic-state distillation

A logical gate is *single-qubit transversal* if, on the code space,

$$\bar{U} = U_1 \otimes \cdots \otimes U_n.$$

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.

[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and magic-state distillation

A logical gate is *single-qubit transversal* if, on the code space,

$$\bar{U} = U_1 \otimes \cdots \otimes U_n.$$

Eastin–Knill [EK09]: no code that detects every single-qubit error has a universal transversal gate set.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and magic-state distillation

A logical gate is *single-qubit transversal* if, on the code space,

$$\bar{U} = U_1 \otimes \cdots \otimes U_n.$$

Eastin–Knill [EK09]: no code that detects every single-qubit error has a universal transversal gate set.

Magic-state distillation [BK05] uses Clifford operations:

$$\text{many noisy } T|+\rangle \text{ or } \mathrm{CCZ}|{+}{+}{+}\rangle \longrightarrow \text{ fewer, less noisy copies.}$$

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and magic-state distillation

A logical gate is *single-qubit transversal* if, on the code space,

$$\bar{U} = U_1 \otimes \cdots \otimes U_n.$$

Eastin–Knill [EK09]: no code that detects every single-qubit error has a universal transversal gate set.

Magic-state distillation [BK05] uses Clifford operations:

$$\text{many noisy } T|+\rangle \text{ or } \mathrm{CCZ}|{+}{+}{+}\rangle \longrightarrow \text{ fewer, less noisy copies.}$$

Most distillation protocols need a stabilizer code with a transversal logical T or CCZ. This talk finds such codes.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

$\mathrm{RM}(r,m)$: Reed–Muller code, degree $\leq r$ in m variables. Punctured: one coordinate deleted. $\mathbf{1}$: all-ones vector.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

$\mathrm{RM}(r,m)$: Reed–Muller code, degree $\leq r$ in m variables. Punctured: one coordinate deleted. $\mathbf{1}$: all-ones vector.

Every CSS code has transversal CNOT, so [EK09] rules out transversal T for Steane and H for $[\![15,1,3]\!]$.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

$\mathrm{RM}(r,m)$: Reed–Muller code, degree $\leq r$ in m variables. Punctured: one coordinate deleted. $\mathbf{1}$: all-ones vector.

Every CSS code has transversal CNOT, so [EK09] rules out transversal T for Steane and H for $[\![15,1,3]\!]$.

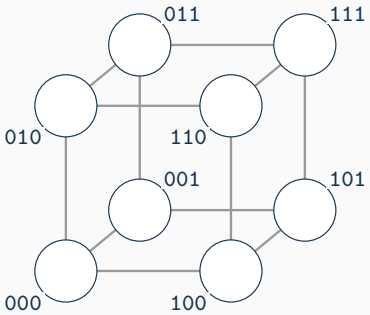
Which CSS codes with $k = 3$ have $U_j \in \{T, T^\dagger\}$ with $U_1 \otimes \cdots \otimes U_n$ logical CCZ?

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Outline

1. Quantum error correction and transversal gates
2. **The cube code: a worked example of logical CCZ**
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

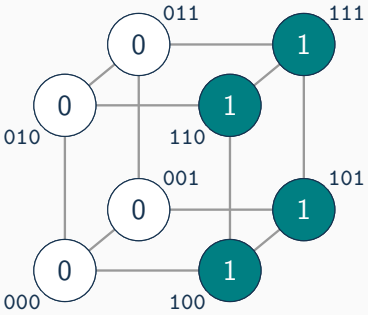
Qubits indexed by the points of $\mathbb{F}_2^3$



Index the qubits by the points $u \in \mathbb{F}_2^3$, so a binary vector is a Boolean function:

$$c \colon \mathbb{F}_2^3 \to \mathbb{F}_2.$$

Qubits indexed by the points of $\mathbb{F}_2^3$



The vector u_1.

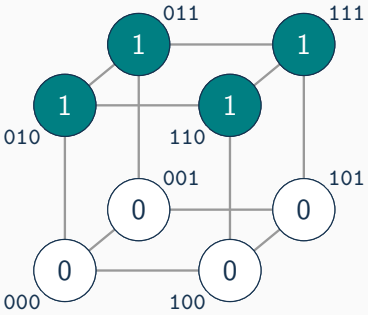
Index the qubits by the points $u \in \mathbb{F}_2^3$, so a binary vector is a Boolean function:

$$c\colon \mathbb{F}_2^3 \to \mathbb{F}_2.$$

Polynomials in u_1, u_2, u_3 give binary vectors, in the order $000, 001, \ldots, 111$:

$$u_1 = 00001111.$$

Qubits indexed by the points of $\mathbb{F}_2^3$



The vector u_2.

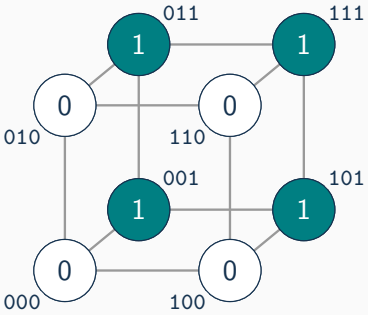
Index the qubits by the points $u \in \mathbb{F}_2^3$, so a binary vector is a Boolean function:

$$c\colon \mathbb{F}_2^3 \to \mathbb{F}_2.$$

Polynomials in u_1, u_2, u_3 give binary vectors, in the order $000, 001, \ldots, 111$:

$$u_1 = 00001111.$$

Qubits indexed by the points of $\mathbb{F}_2^3$



The vector u_3.

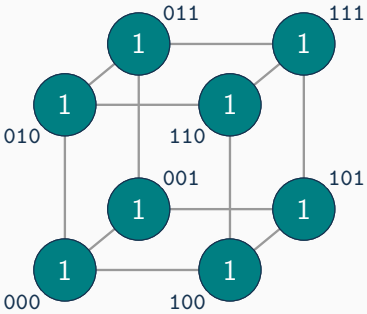
Index the qubits by the points $u \in \mathbb{F}_2^3$, so a binary vector is a Boolean function:

$$c\colon \mathbb{F}_2^3 \to \mathbb{F}_2.$$

Polynomials in u_1, u_2, u_3 give binary vectors, in the order $000, 001, \ldots, 111$:

$$u_1 = 00001111.$$

Qubits indexed by the points of $\mathbb{F}_2^3$



The constant vector $\mathbf{1}$.

Index the qubits by the points $u \in \mathbb{F}_2^3$, so a binary vector is a Boolean function:

$$c\colon \mathbb{F}_2^3 \to \mathbb{F}_2.$$

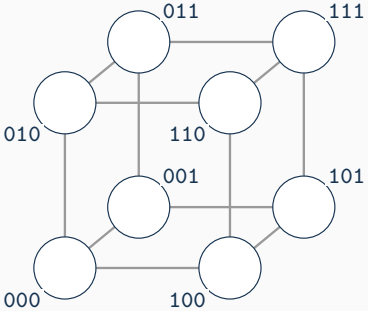
Polynomials in u_1, u_2, u_3 give binary vectors, in the order $000, 001, \ldots, 111$:

$$u_1 = 00001111.$$

Degree at most 1:

$$\mathrm{RM}(1,3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}, \quad [8,4,4].$$

The cube code $[\![8, 3, 2]\!]$



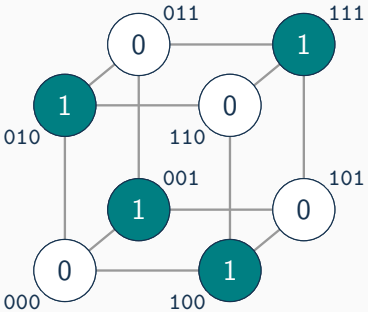
One X-type generator $X(\mathbf{1})$ and three logical rows:

$$S = \mathbf{1}, \qquad K = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \qquad \mathcal{C}_1 = \mathrm{RM}(1, 3).$$

With $K^\top x = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$:

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

The cube code $[\![8, 3, 2]\!]$



$x \cdot u = u_1 \oplus u_2 \oplus u_3$ for $x = 111$.

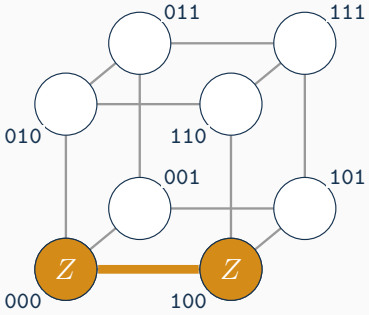
One X-type generator $X(\mathbf{1})$ and three logical rows:

$$S = \mathbf{1}, \qquad K = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \qquad \mathcal{C}_1 = \mathrm{RM}(1, 3).$$

With $K^\top x = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$:

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

Distances of the cube code

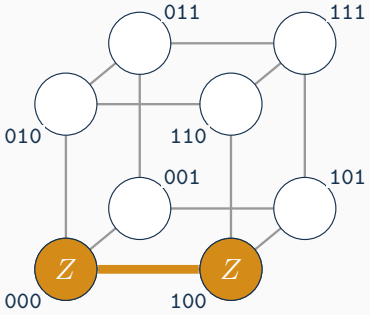


$Z(e)$, $e = \delta_{000} \oplus \delta_{100}$.

Z on the qubits 000 and 100, with δ_u the indicator of u:

$$Z(e), \qquad e = \delta_{000} \oplus \delta_{100}.$$

Distances of the cube code



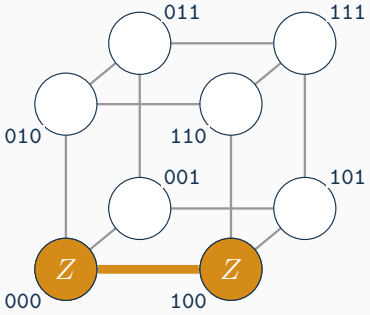
$Z(e)$, $e = \delta_{000} \oplus \delta_{100}$.

Z on the qubits 000 and 100, with δ_u the indicator of u:

$$Z(e), \qquad e = \delta_{000} \oplus \delta_{100}.$$

- $e \cdot \mathbf{1} = 0$: commutes with $X(\mathbf{1})$, so $d_Z \geq 2$.

Distances of the cube code



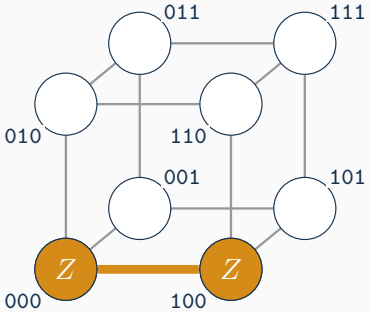
$Z(e)$, $e = \delta_{000} \oplus \delta_{100}$.

Z on the qubits 000 and 100, with δ_u the indicator of u:

$$Z(e), \qquad e = \delta_{000} \oplus \delta_{100}.$$

- $e \cdot \mathbf{1} = 0$: commutes with $X(\mathbf{1})$, so $d_Z \geq 2$.
- $Z(e)|x\rangle_L = (-1)^{x_1}|x\rangle_L$: a logical operator, so $d_Z = 2$.

Distances of the cube code



$Z(e)$, $e = \delta_{000} \oplus \delta_{100}$.

Z on the qubits 000 and 100, with δ_u the indicator of u:

$$Z(e), \qquad e = \delta_{000} \oplus \delta_{100}.$$

- $e \cdot \mathbf{1} = 0$: commutes with $X(\mathbf{1})$, so $d_Z \geq 2$.
- $Z(e)|x\rangle_L = (-1)^{x_1}|x\rangle_L$: a logical operator, so $d_Z = 2$.
- $d_X = 4$, the minimum weight of $\mathrm{RM}(1,3)$.

Codeword weight, physical phase, and logical phase

A $T/T^\dagger$ *pattern* puts T^{Γ_j} on qubit j, with $\Gamma \in \{\pm 1\}^n$ and the *signed weight* sw_Γ:

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi\,\mathrm{sw}_\Gamma(c)/4}|c\rangle, \qquad \mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j.$$

Codeword weight, physical phase, and logical phase

A $T/T^\dagger$ *pattern* puts T^{Γ_j} on qubit j, with $\Gamma \in \{\pm 1\}^n$ and the *signed weight* sw_Γ:

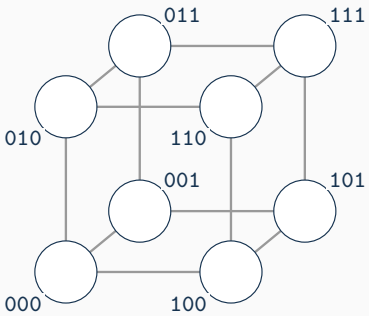
$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi\,\mathrm{sw}_\Gamma(c)/4}|c\rangle, \qquad \mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j.$$

Logical CCZ: every codeword of the coset $K^\top x \oplus \mathcal{C}_2$ gets the phase $(-1)^{x_1x_2x_3}$:

$$\mathrm{sw}_\Gamma(K^\top x \oplus S^\top y) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

The gate condition is a congruence on integer signed weights.

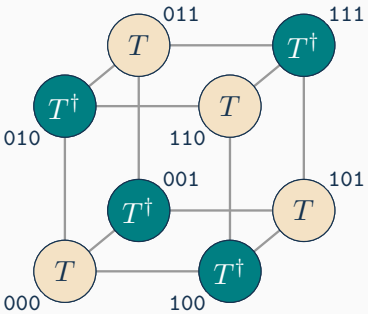
Choosing the signs on the cube



Codewords $c = x \cdot u \oplus y\mathbf{1}$ are affine, with 4 ones unless constant. The codeword of $x = 111$:

$$\gamma = u_1 \oplus u_2 \oplus u_3.$$

Choosing the signs on the cube



$T^\dagger$ where $\gamma(u) = 1$ [CH17].

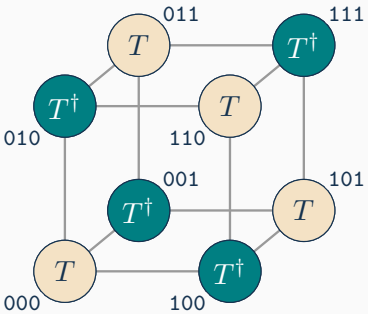
Codewords $c = x \cdot u \oplus y\mathbf{1}$ are affine, with 4 ones unless constant. The codeword of $x = 111$:

$$\gamma = u_1 \oplus u_2 \oplus u_3.$$

Put $T^\dagger$ where $\gamma = 1$, that is $\Gamma_u = (-1)^{\gamma(u)}$:

$$\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c| = |\gamma \oplus c| - 4.$$

Choosing the signs on the cube



$T^\dagger$ where $\gamma(u) = 1$ [CH17].

Codewords $c = x \cdot u \oplus y\mathbf{1}$ are affine, with 4 ones unless constant. The codeword of $x = 111$:

$$\gamma = u_1 \oplus u_2 \oplus u_3.$$

Put $T^\dagger$ where $\gamma = 1$, that is $\Gamma_u = (-1)^{\gamma(u)}$:

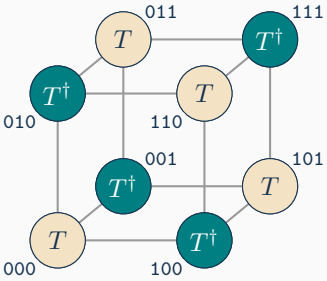
$$\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c| = |\gamma \oplus c| - 4.$$

$\gamma \oplus c$ is nonconstant unless $x = 111$:

$$|\gamma \oplus c| = \begin{cases} 4, & x \neq 111, \\ 0 \text{ or } 8, & x = 111. \end{cases}$$

$\mathrm{sw}_\Gamma(c) \equiv 4x_1x_2x_3 \pmod 8$: logical CCZ.

Reading the cube phase from overlaps



XOR of bits as an integer:

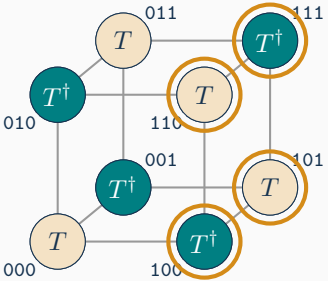
$$a_1 \oplus a_2 \oplus a_3 = \textstyle\sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

Applied at each vertex to $c = K^\top x$, weighted by Γ_u:

$$\mathrm{sw}_\Gamma(c) = \textstyle\sum_i x_i\,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_i x_j\,\mathrm{sw}_\Gamma(K_i \wedge K_j) + 4x_1x_2x_3\;\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

sw_Γ :

Reading the cube phase from overlaps



$K_1 = u_1$: the face $u_1 = 1$.

XOR of bits as an integer:

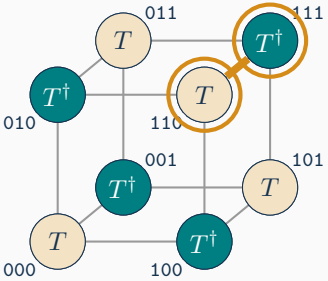
$$a_1 \oplus a_2 \oplus a_3 = \sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

Applied at each vertex to $c = K^\top x$, weighted by Γ_u:

$$\mathrm{sw}_\Gamma(c) = \sum_i x_i \,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_i x_j \,\mathrm{sw}_\Gamma(K_i \wedge K_j) + 4x_1x_2x_3 \,\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

sw_Γ : face 0,

Reading the cube phase from overlaps



$K_1 \wedge K_2 = u_1 u_2.$

XOR of bits as an integer:

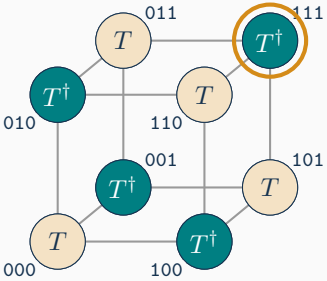
$$a_1 \oplus a_2 \oplus a_3 = \textstyle\sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

Applied at each vertex to $c = K^\top x$, weighted by Γ_u:

$$\mathrm{sw}_\Gamma(c) = \textstyle\sum_i x_i\,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_i x_j\,\mathrm{sw}_\Gamma(K_i \wedge K_j) + 4x_1x_2x_3\;\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

sw_Γ : face 0, edge 0,

Reading the cube phase from overlaps



$K_1 \wedge K_2 \wedge K_3 = u_1 u_2 u_3.$

XOR of bits as an integer:

$$a_1 \oplus a_2 \oplus a_3 = \textstyle\sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

Applied at each vertex to $c = K^\top x$, weighted by Γ_u:

$$\mathrm{sw}_\Gamma(c) = \textstyle\sum_i x_i\, \mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_i x_j\, \mathrm{sw}_\Gamma(K_i \wedge K_j) + 4x_1x_2x_3\ \mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

sw_Γ : face 0, edge 0, vertex 111: -1.

$\mathrm{sw}_\Gamma(c) = -4x_1x_2x_3 \equiv 4x_1x_2x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

The *Bravyi–Haah parity identity* [BH12], for any binary vectors:

$$\mathrm{sw}_\Gamma(a_1 \oplus \cdots \oplus a_t) = \textstyle\sum_{J\neq\emptyset}(-2)^{|J|-1}\ \mathrm{sw}_\Gamma\big(\bigwedge_{j\in J} a_j\big).$$

Comparing coefficients in the gate condition gives, for distinct rows g, g', g'' of G:

$$\begin{aligned}
\mathrm{sw}_\Gamma(g) &\equiv 0 \pmod 8,\\
\mathrm{sw}_\Gamma(g \wedge g') &\equiv 0 \pmod 4,\\
\mathrm{sw}_\Gamma(g \wedge g' \wedge g'') &\equiv \begin{cases} 1, & \{g,g',g''\} = \{K_1,K_2,K_3\},\\ 0, & \text{otherwise,}\end{cases} \pmod 2.
\end{aligned}$$

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

The gate condition becomes conditions on generator rows

The *Bravyi–Haah parity identity* [BH12], for any binary vectors:

$$\mathrm{sw}_\Gamma(a_1 \oplus \cdots \oplus a_t) = \textstyle\sum_{J \neq \emptyset} (-2)^{|J|-1}\, \mathrm{sw}_\Gamma\big(\bigwedge_{j \in J} a_j\big).$$

Comparing coefficients in the gate condition gives, for distinct rows g, g', g'' of G:

$$\begin{aligned}
\mathrm{sw}_\Gamma(g) &\equiv 0 \pmod 8,\\
\mathrm{sw}_\Gamma(g \wedge g') &\equiv 0 \pmod 4,\\
\mathrm{sw}_\Gamma(g \wedge g' \wedge g'') &\equiv \begin{cases} 1, & \{g, g', g''\} = \{K_1, K_2, K_3\},\\ 0, & \text{otherwise,} \end{cases} \pmod 2.
\end{aligned}$$

The cube code satisfies all three: faces and edges give 0, and the vertex 111 gives -1.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. **Triorthogonality and the CCZ conditions**
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Historical context

2005 Bravyi–Kitaev [BK05]: magic-state distillation.

2012 Bravyi–Haah [BH12]: triorthogonal matrices for transversal T.

2017 Campbell–Howard [CH17]: CCZ and other multiqubit gates.

2022 Nezami–Haah [NH22]: triorthogonal codes with $n + k \leq 38$ classified via Reed–Muller codes.

2024 Camps-Moreno et al. [CLM+24]: CSS-T codes, on which $T^{\otimes n}$ is logical. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

Triorthogonal matrix and triorthogonal space

Matrix G **[BH12]**, distinct rows:

$$|g \wedge g'| \equiv |g \wedge g' \wedge g''| \equiv 0 \pmod 2.$$

Odd rows give logical X, even rows give X-type stabilizers.

Space $\mathcal{H} \subseteq \mathbb{F}_2^L$ **[NH22]**, repeats allowed:

$$\sum_j h_j h'_j h''_j \equiv 0 \pmod 2 \quad \forall\, h, h', h'' \in \mathcal{H}.$$

Unital: $\mathbf{1} \in \mathcal{H}$. Every weight is even.

Triorthogonal matrix and triorthogonal space

Matrix G **[BH12]**, distinct rows:

$$|g \wedge g'| \equiv |g \wedge g' \wedge g''| \equiv 0 \pmod 2.$$

Odd rows give logical X, even rows give X-type stabilizers.

Space $\mathcal{H} \subseteq \mathbb{F}_2^L$ **[NH22]**, repeats allowed:

$$\sum_j h_j h'_j h''_j \equiv 0 \pmod 2 \quad \forall\, h, h', h'' \in \mathcal{H}.$$

Unital: $\mathbf{1} \in \mathcal{H}$. Every weight is even.

[NH22] classify the spaces $\mathcal{H}$, then obtain codes by puncturing.

Triorthogonal matrix and triorthogonal space

Matrix G [BH12], distinct rows:

$$|g \wedge g'| \equiv |g \wedge g' \wedge g''| \equiv 0 \pmod 2.$$

Odd rows give logical X, even rows give X-type stabilizers.

Space $\mathcal{H} \subseteq \mathbb{F}_2^L$ [NH22], repeats allowed:

$$\sum_j h_j h'_j h''_j \equiv 0 \pmod 2 \quad \forall\, h, h', h'' \in \mathcal{H}.$$

Unital: $\mathbf{1} \in \mathcal{H}$. Every weight is even.

[NH22] classify the spaces $\mathcal{H}$, then obtain codes by puncturing.

We fix S by geometry, then search for K. CCZ needs one odd triple, $K_1 \wedge K_2 \wedge K_3$, which neither definition allows.

Conditions for quasi-transversal CCZ

K1	$\lvert K_i\rvert \equiv 0$
K2	$\lvert K_i \wedge K_{i'}\rvert \equiv 0$
K3	$\lvert K_1 \wedge K_2 \wedge K_3\rvert \equiv 1$
KS1	$\lvert K_i \wedge S_l\rvert \equiv 0$
KS2	$\lvert K_i \wedge K_{i'} \wedge S_l\rvert \equiv 0$
KS3	$\lvert K_i \wedge S_l \wedge S_{l'}\rvert \equiv 0$
S1	$\lvert S_l\rvert \equiv 0$
S2	$\lvert S_l \wedge S_{l'}\rvert \equiv 0$
S3	$\lvert S_l \wedge S_{l'} \wedge S_{l''}\rvert \equiv 0$

Rows K_i of K, S_l of S. All mod 2, indices distinct.

The three row conditions reduce mod 2 to the nine Campbell–Howard (CH) conditions [CH17]:

$$\Gamma_j \text{ odd } \Rightarrow \ \mathrm{sw}_\Gamma(c) \equiv |c| \pmod 2.$$

Conditions for quasi-transversal CCZ

K1	$\lvert K_i\rvert \equiv 0$
K2	$\lvert K_i \wedge K_{i'}\rvert \equiv 0$
K3	$\lvert K_1 \wedge K_2 \wedge K_3\rvert \equiv 1$
KS1	$\lvert K_i \wedge S_l\rvert \equiv 0$
KS2	$\lvert K_i \wedge K_{i'} \wedge S_l\rvert \equiv 0$
KS3	$\lvert K_i \wedge S_l \wedge S_{l'}\rvert \equiv 0$
S1	$\lvert S_l\rvert \equiv 0$
S2	$\lvert S_l \wedge S_{l'}\rvert \equiv 0$
S3	$\lvert S_l \wedge S_{l'} \wedge S_{l''}\rvert \equiv 0$

Rows K_i of K, S_l of S. All mod 2, indices distinct.

The three row conditions reduce mod 2 to the nine Campbell–Howard (CH) conditions [CH17]:

$$\Gamma_j \text{ odd } \Rightarrow \ \mathrm{sw}_\Gamma(c) \equiv |c| \pmod 2.$$

They give *quasi-transversal* CCZ, with a diagonal Clifford *correction* C:

$$C\,T^{\otimes n} = \overline{\mathrm{CCZ}}.$$

Conditions for quasi-transversal CCZ

K1	$\lvert K_i\rvert \equiv 0$
K2	$\lvert K_i \wedge K_{i'}\rvert \equiv 0$
K3	$\lvert K_1 \wedge K_2 \wedge K_3\rvert \equiv 1$
KS1	$\lvert K_i \wedge S_l\rvert \equiv 0$
KS2	$\lvert K_i \wedge K_{i'} \wedge S_l\rvert \equiv 0$
KS3	$\lvert K_i \wedge S_l \wedge S_{l'}\rvert \equiv 0$
S1	$\lvert S_l\rvert \equiv 0$
S2	$\lvert S_l \wedge S_{l'}\rvert \equiv 0$
S3	$\lvert S_l \wedge S_{l'} \wedge S_{l''}\rvert \equiv 0$

Rows K_i of K, S_l of S. All mod 2, indices distinct.

The three row conditions reduce mod 2 to the nine Campbell–Howard (CH) conditions [CH17]:

$$\Gamma_j \text{ odd } \Rightarrow \ \mathrm{sw}_\Gamma(c) \equiv |c| \pmod 2.$$

They give *quasi-transversal* CCZ, with a diagonal Clifford *correction* C:

$$C\,T^{\otimes n} = \overline{\mathrm{CCZ}}.$$

A $T/T^\dagger$ pattern with no correction needs the mod 8 and mod 4 conditions on Γ.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. **Projective geometry and the 48-qubit code**
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$\mathcal{Q}_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A $T/T^\dagger$ pattern with $26\,T$ and $22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- $\mathcal{C}_2$ is fixed by projective geometry: the columns of S are the points of $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$, the nonzero vectors of $\mathbb{F}_2^6$ outside a 4-dimensional subspace.
- The logical rows K come from a SAT (Boolean satisfiability) search.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- Removing other subspaces gives codes at 13 block lengths from 48 to 496.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.

2. $\mathcal{C}_2$ is *8-divisible*: every codeword has weight $\equiv 0 \pmod 8$. By the Bravyi–Haah parity identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. $\mathcal{C}_2$ is *8-divisible*: every codeword has weight $\equiv 0 \pmod 8$. By the Bravyi–Haah parity identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. $\mathcal{C}_2$ is *8-divisible*: every codeword has weight $\equiv 0 \pmod 8$. By the Bravyi–Haah parity identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.
4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error on qubit j has syndrome β_j, and Z errors on qubits $i \neq j$ have syndrome $\beta_i \oplus \beta_j$.

- $\beta_j \neq 0$ for all j: every single Z error is detected.
- $\beta_i \neq \beta_j$ for all $i \neq j$: every pair of Z errors is detected.

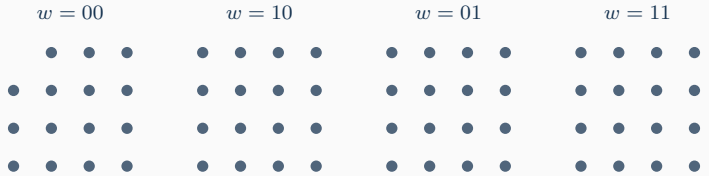
Such S is called *projective*: its columns are distinct points of $\mathrm{PG}(s-1, 2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the columns of S alone, for every choice of logical rows K.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

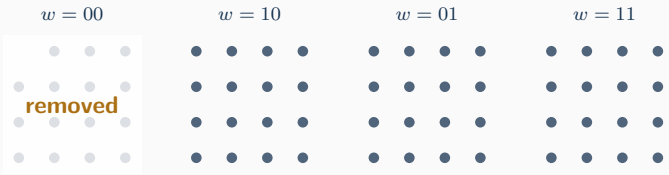
The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$.



Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$.

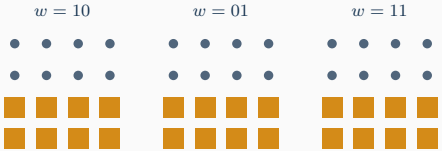


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ code $\mathcal{C}_2$.

Hyperplane sections and codeword weights

Each codeword c of $\mathcal{C}_2$ is a linear function $c(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $|c| = 48 - |\mathcal{P} \cap \{c = 0\}|$.



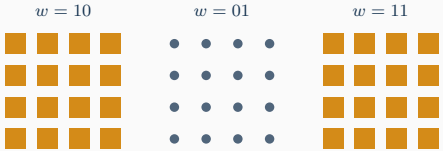
$\lambda \neq 0$, for example $c = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

Hyperplane sections and codeword weights

Each codeword c of $\mathcal{C}_2$ is a linear function $c(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $|c| = 48 - |\mathcal{P} \cap \{c = 0\}|$.



$\lambda = 0$, for example $c = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q-1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p\backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no $T/T^\dagger$ implementation of CCZ.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. **Column indicators, logical rows, and phase signs**
6. Lower bounds and open questions

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, with x for the rows of K and y for the row of S, the columns are exactly the points with $y = 1$, so $\chi_G(x, y) = y$.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0&0&0&0&1&1&1&1 \\ 0&0&1&1&0&0&1&1 \\ 0&1&0&1&0&1&0&1 \\ 1&1&1&1&1&1&1&1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, with x for the rows of K and y for the row of S, the columns are exactly the points with $y = 1$, so $\chi_G(x, y) = y$.

$\mathcal{Q}_{48}$: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, f' \rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set J of rows, $m_J = \prod_{j \in J} z_j$ and

$$\langle \chi_G, m_J \rangle = \left| \bigwedge_{j \in J} g_j \right| \bmod 2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, f' \rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set J of rows, $m_J = \prod_{j \in J} z_j$ and

$$\langle \chi_G, m_J \rangle = \left| \bigwedge_{j \in J} g_j \right| \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1 x_2 x_3 \rangle = 1$, and $\langle \chi_G, m \rangle = 0$ for every other monomial of degree 1 to 3.

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, f' \rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set J of rows, $m_J = \prod_{j \in J} z_j$ and

$$\langle \chi_G, m_J \rangle = \Big| \bigwedge_{j \in J} g_j \Big| \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1 x_2 x_3 \rangle = 1$, and $\langle \chi_G, m \rangle = 0$ for every other monomial of degree 1 to 3.

Since $\mathrm{RM}(3, r)^{\perp} = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) satisfies these conditions [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon \delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2,$$

where δ_0 is the indicator of the origin.

Logical rows in the column picture

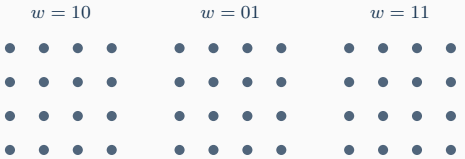
Column $\beta = (u, w) \in \mathcal{P}$ of S has the *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set J of rows into its logical rows J_K and stabilizer rows J_S. The CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{J_K}(\kappa(\beta))\, m_{J_S}(\beta) = \begin{cases} 1, & J = \{K_1, K_2, K_3\}, \\ 0, & \text{else.} \end{cases}$$

S1–S3 have $J_K = \emptyset$ and hold because $\mathcal{C}_2$ is 8-divisible. The rest have degree at most 3 in the label bits.



Logical rows in the column picture

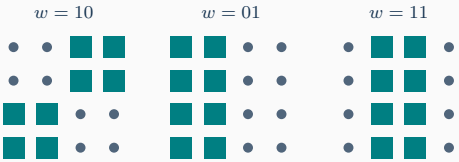
Column $\beta = (u, w) \in \mathcal{P}$ of S has the *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set J of rows into its logical rows J_K and stabilizer rows J_S. The CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{J_K}(\kappa(\beta))\, m_{J_S}(\beta) = \begin{cases} 1, & J = \{K_1, K_2, K_3\}, \\ 0, & \text{else.} \end{cases}$$

S1–S3 have $J_K = \emptyset$ and hold because $\mathcal{C}_2$ is 8-divisible. The rest have degree at most 3 in the label bits.



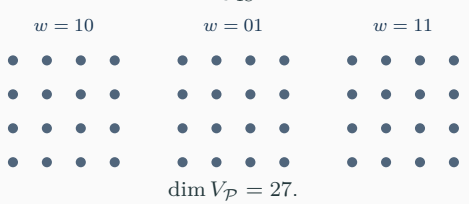
Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 + u_3$, $1 + u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Row K_i is $\kappa_i \colon \mathcal{P} \to \mathbb{F}_2$, bit i of the label map, and $\langle f, f' \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) f'(\beta) \bmod 2$ [LB26].

1. K1, KS1, KS3 say $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ (y_l: row l of S), which span $\mathrm{RM}(2, s)$:

$$\kappa_i \in V_{\mathcal{P}} = \{ f : \langle f, m \rangle_{\mathcal{P}} = 0,\ m \in \mathrm{RM}(2, s) \}.$$



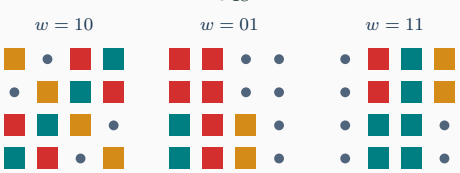
Finding the logical rows

Fix $\mathcal{P}$. Row K_i is $\kappa_i \colon \mathcal{P} \to \mathbb{F}_2$, bit i of the label map, and $\langle f, f' \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) f'(\beta) \bmod 2$ [LB26].

1. K1, KS1, KS3 say $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ (y_l: row l of S), which span $\mathrm{RM}(2, s)$:

$$\kappa_i \in V_{\mathcal{P}} = \{ f : \langle f, m \rangle_{\mathcal{P}} = 0,\ m \in \mathrm{RM}(2, s) \}.$$

2. K2, KS2: choose a pair with $\langle \kappa_1 \kappa_2, \mathbf{1} \rangle_{\mathcal{P}} = \langle \kappa_1 \kappa_2, y_l \rangle_{\mathcal{P}} = 0$.



Teal: only K_1 has a 1. Gold: only K_2 has a 1. Red: both. $|K_1 \wedge K_2| = 12 \equiv 0 \pmod 2$ (K2).

Finding the logical rows

Fix $\mathcal{P}$. Row K_i is $\kappa_i \colon \mathcal{P} \to \mathbb{F}_2$, bit i of the label map, and $\langle f, f' \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) f'(\beta) \bmod 2$ [LB26].

1. K1, KS1, KS3 say $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ (y_l: row l of S), which span $\mathrm{RM}(2, s)$:

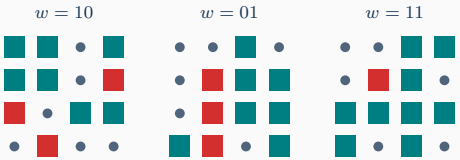
$$\kappa_i \in V_{\mathcal{P}} = \{ f : \langle f, m \rangle_{\mathcal{P}} = 0,\ m \in \mathrm{RM}(2, s) \}.$$

2. K2, KS2: choose a pair with $\langle \kappa_1 \kappa_2, \mathbf{1} \rangle_{\mathcal{P}} = \langle \kappa_1 \kappa_2, y_l \rangle_{\mathcal{P}} = 0$.

3. The rest is linear in $\kappa_3 \in V_{\mathcal{P}}$, solved by Gaussian elimination:

$$\langle \kappa_3, \kappa_i \rangle_{\mathcal{P}} = \langle \kappa_3, y_l \kappa_i \rangle_{\mathcal{P}} = 0,\ \langle \kappa_3, \kappa_1 \kappa_2 \rangle_{\mathcal{P}} = 1.$$

$\mathcal{Q}_{48}$



Squares: K_3 has a 1. Red: K_1 and K_2 too.
$|K_1 \wedge K_2 \wedge K_3| = 7 \equiv 1 \pmod 2$ (K3).

Finding the logical rows

Fix $\mathcal{P}$. Row K_i is $\kappa_i \colon \mathcal{P} \to \mathbb{F}_2$, bit i of the label map, and $\langle f, f' \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) f'(\beta) \bmod 2$ [LB26].

1. K1, KS1, KS3 say $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ (y_l: row l of S), which span $\mathrm{RM}(2, s)$:

   $$\kappa_i \in V_{\mathcal{P}} = \{ f : \langle f, m \rangle_{\mathcal{P}} = 0,\ m \in \mathrm{RM}(2, s) \}.$$

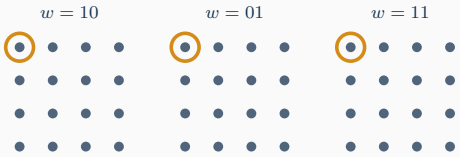
2. K2, KS2: choose a pair with $\langle \kappa_1 \kappa_2, \mathbf{1} \rangle_{\mathcal{P}} = \langle \kappa_1 \kappa_2, y_l \rangle_{\mathcal{P}} = 0$.

3. The rest is linear in $\kappa_3 \in V_{\mathcal{P}}$, solved by Gaussian elimination:

   $$\langle \kappa_3, \kappa_i \rangle_{\mathcal{P}} = \langle \kappa_3, y_l \kappa_i \rangle_{\mathcal{P}} = 0,\ \langle \kappa_3, \kappa_1 \kappa_2 \rangle_{\mathcal{P}} = 1.$$

   If no pair extends, $\mathcal{P}$ is excluded.

$\mathcal{Q}_{48}$

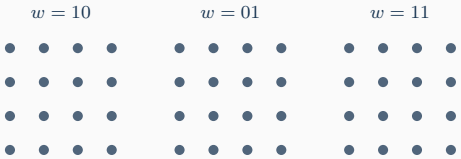


Z at $u = 0000$: labels $011 \oplus 110 \oplus 000 = 101$, so $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

As on the cube, $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j = 1$ on the $T^\dagger$ qubits. Then $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c|$, and every codeword $c = K^\top x \oplus S^\top y$ has even weight:

$$|\gamma \wedge c| \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4.$$



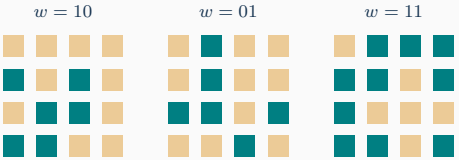
Solving for the $T/T^\dagger$ pattern

As on the cube, $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j = 1$ on the $T^\dagger$ qubits. Then $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c|$, and every codeword $c = K^\top x \oplus S^\top y$ has even weight:

$$|\gamma \wedge c| \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4.$$

For $\mathcal{Q}_{48}$ [LB26]:

$$\begin{aligned}\gamma(u,w) &= u_3 + w_1Q_1(u) + w_2Q_2(u),\\ Q_1 &= u_1 + u_2 + u_3 + u_1u_3 + u_2u_3 + u_2u_4,\\ Q_2 &= u_3 + u_4 + u_1u_2 + u_1u_4 + u_3u_4.\end{aligned}$$



Teal: $T^\dagger$ $(6, 6, 10$ per block). Gold: T. Total $26\,T + 22\,T^\dagger$.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. **Lower bounds and open questions**

Stabilizer supports are Reed–Muller codewords

For projective S, S1–S3 say $\langle \chi_S, m \rangle = 0$ for every monomial m in $y_1, \ldots, y_s$ of degree 1 to 3.

Following [NH22], let $c = \chi_S + \varepsilon \delta_0$, where ε makes $|c|$ even. Every monomial of degree at least 1 vanishes at 0, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = |c| \bmod 2$. Hence $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = |c| - c(0).$$

For $\mathcal{Q}_{48}$: $c = \chi_S = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce the stabilizer rank s (the number of rows of S) to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the possible weights and the geometry of the support.
3. At rank 8, use the classification of low-weight Reed–Muller codewords to reduce to finitely many supports, and enumerate them.
4. Show that no remaining support admits logical rows satisfying K1–K3 and KS1–KS3.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify generalized triorthogonal codes with $n \leq 54$ and $d_Z \geq 3$, which covers quasi-transversal CCZ with $k = 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no $T/T^\dagger$ pattern [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat (a translate of a 4-dimensional subspace), with the origin deleted. $\mathcal{Q}_{48}$ removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with a circuit footprint of 9 qubits.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $\mathcal{Q}_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Our bound excludes $n \leq 38$, and [WJS26] excludes 39–46.

Open questions

- Which supports admit logical rows K and signs Γ?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public